Subject Description Form

Subject Code	FSN1001
Subject Title	Essential Laboratory Practices
Credit Value	1
Level	1
Pre-requisite	NIL
Objectives	To introduce the common techniques and safety practices required in laboratories.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Demonstrate the familiarity towards common laboratory techniques; b. Compile proper laboratory records from the observations made in experiments; c. Observe general laboratory safety practices; d. Pass the online safety training for both Chemical and Biological Safety.
Subject Synopsis/ Indicative Syllabus	**Laboratory Safety** - General laboratory safety practices. - Equipment designed to reduce biological hazards. - General knowledge on the handling, storage and disposal of chemicals and chemical wastes. - Personal protection and protective clothing. - Use of emergency facilities. **Essential Laboratory Techniques** - Use of pH meter, analytical balances, graduated glassware; water for laboratory use; concentrations and calculation; preparation of laboratory solutions, reagents and standard solutions. Use of auto-pipette - Acid-base titration - Measurements involving light: transmission, absorption, principle of spectrophotometry, use of spectrophotometer; standard curves and calibration. - Record keeping and documentation; essential data analysis and report writing

Teaching/Learning Methodology

The basic principles of laboratory safety will be delivered in the form of lectures. To practice, students will work individually or in teams in the laboratory sessions, and each session will be supplemented with in-lab briefing and demonstration.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomesto be assessed (Please tick as appropriate)				
		a	b	c	d	
1. Lab reports	40%	√		√		
3. Lab Performances	30%	√	√	√		
4. Quiz	30%				√	
Total	100 %					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Laboratory reports: students are expected to keep record of experimental data, to perform analysis on the data obtained, and to interpret their findings. Their abilities in these aspects may thus be assessed.

Laboratory performance: students will be supervised during the laboratory sessions and their laboratory performance, including mastering of the basic techniques and practice of laboratory safety, will be assessed.

Quiz: students will be assessed on their understanding of the principles of laboratory techniques and laboratory safety

Student Study Effort Required

Class contact:	
Lecture	6 Hrs.
Laboratory Sessions	9 Hrs.
Other student study effort:	
Self study	20 Hrs.
Total Study Effort:	35 Hrs

Reading List and References

Laboratory Biosafety Manual, Second Edition (Revised); World Health Organization, Geneva 2003

Fleming & Hunt (Editors) Biological Safety Principles and Practices 4th Edition ASM Press 2006

Hall, Stephen K.; Chemical Safety in the Laboratory; Boca Raton, Fla.: Lewis Pubishers, 1994

Quantitative Chemical Analysis, 6th edition, Harlow: Prentice Hall, 2000